Lorenzo Rizzi

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EDUCATION

École Normale Supérieure (ENS - PSL) *Feb 2026 - Present*
Visiting Student at CSD (Centre Sciences des Données, Group of Prof. Loureiro) *Paris, France*

- Building on the group's existing work, investigating random feature models and their **kernel regime limit**, specifically how a **power-law covariate** structure in the training set affects the spectral properties of the kernel operator.
- From the kernel spectrum, deriving **analytical laws for the generalization error** of kernel ridge regression to characterize learning performance in high-dimensional settings, providing an exact theoretical description of phenomena such as **double descent** and **benign overfitting**.
- Audited **M2 Graduate Courses** @ ENS: *Statistical Physics of Disordered Systems* (Profs. V. Ros & A. Rosso), *From Statistical Physics to Machine Learning and Back* (Profs. G. Biroli & M. Gabrié).

University of Padova *Sep 2024 - Present*
M.Sc. in Physics of Data *Padova, Italy*

- GPA: **29.93/30**
- **Core curriculum:** Advanced training combining Physics with Machine Learning, Data Science and Bayesian Statistics.
- **Key Courses:** *Statistical Mechanics of Complex Systems, Numerical Methods for Soft Matter, Machine Learning, Computational Physics Lab, Advanced Bayesian Statistics, Complex Networks, Modern Computing for Physics.*

Scuola Galileiana di Studi Superiori (Galilean School of Higher Studies) *Sep 2024 - Present*
Honours Programme of Excellence of the University of Padova *Padova, Italy*

- **Selection:** Admission based on a highly competitive nationwide exam (merit-based).
- **Award:** Full scholarship, free accommodation and personal academic tutor.
- **Curriculum:** Advanced disciplinary training via dedicated seminars and technical courses (*Path integral in Quantum Mechanics, Introduction to Scientific Computing, Topological states in Condensed Matter*)

University of Bologna *Sep 2021 - Sep 2024*
B.Sc. in Physics *Bologna, Italy*

- Final Grade: **110/110 cum laude**. Final GPA: **29.67/30**.
- **Thesis Project:** Developed a C++ agent-based model to investigate the **phase transition** from free-flow to gridlock in urban networks, analyzing critical behavior under an Origin-Destination paradigm.
- **Resources:** Read Thesis (IT) · View Code (EN)

Collegio Superiore di Bologna *Sep 2021 - Sep 2024*
School of Excellence of the University of Bologna *Bologna, Italy*

- **Overview:** Merit-based Honours College offering high-level interdisciplinary training parallel to standard university courses.
- **Selection:** Admission based on a highly competitive entrance examination.

SKILLS AND LANGUAGES

Programming	C/C++ & OOP, Python (NumPy, Pandas, Dask), R
HPC & Tools	CUDA C (GPU Computing), OpenMP & MPI, Docker, Linux/Bash, Git, LaTeX
Languages	Italian (Native), English (C1), French (C1, *ESABAC Double Diploma*)

SELECTION OF ACADEMIC PROJECTS

Bayesian Analysis for T2K Experiment *2026 - Present*
Research Project under the supervision of Prof. A. Longhin (INFN Padova)

- Studying the architecture of **MaCh3** (Markov Chain 3), the Bayesian fitting framework used by the T2K collaboration to infer fundamental parameters in neutrino physics.
- Investigating the behavior of the MCMC algorithm by isolating specific **neutrino interaction samples** to validate the fitting pipeline or find interesting pattern.

Parallel CUDA and OpenMP N-body problem *2026*
View Code on GitHub

- Developed an OpenMP and **CUDA-accelerated** solver for the gravitational N-body problem, optimizing kernel execution and memory access patterns.

Numerical Methods for Soft Matter *2025*
View Code on GitHub

- Developed a C++ library for Statistical Mechanics simulations from scratch. Implemented **Molecular Dynamics** engines (Verlet and various thermostats) and **MCMC algorithms**.
- Analyzed critical phenomena in 2D Ising models and active matter systems (**Active Ising model**) to study non-equilibrium phase transitions and detailed balance breaking.

Urban Network Phase Transitions *2024*
View Code on GitHub

- Designed an agent-based Origin-Destination model to simulate traffic dynamics on a **quenched urban network**. Agents move according to specific microscopic dynamics, giving rise to global thermodynamic behavior.
- Analyzed macroscopic observables to characterize a **physical phase transition** from a free-flow regime to a state of total congestion (gridlock).

HONORS & AWARDS

- **2024:** Full Scholarship, Scuola Galileiana di Studi Superiori (Padova).
- **2021:** Full Scholarship, Collegio Superiore (Bologna).
- **2021:** Silver Medal @ *Olimpiadi della Fisica*, National Physics Olympics (Italy).

SCHOOLS AND CONFERENCES

GraSPA 2024 Annecy, France · July 2024
Summer School on Particle and Astroparticle Physics (LAPTh). *Selected Participant.*

ACML 2023 Istanbul, Turkey · Nov 2023
Asian Conference on Machine Learning. *Attendee.*

25/02/2026
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